

Topic 1 – Setting Up the n8n Advanced AI Agent Environment

Workflow Documentation

1. Assessment Objective

The objective is to build, configure, execute, and validate a functional n8n AI Agent workflow using secure credential management, model configuration, workflow execution, and execution monitoring.

2. Workflow Overview

The completed workflow uses three components: **When Chat Message Received** → **AI Agent**, with an **OpenAI Chat Model** connected to the AI Agent's language-model input.

Component	Configuration / Purpose
When Chat Message Received	Receives the user's chat message and passes it to the AI Agent.
AI Agent	Processes the incoming message using the connected chat model. System message: "You are a helpful a
OpenAI Chat Model	OpenAI GPT-4.1; credential: Gateway credits; Temperature: 0.0; Max Tokens: 500.

3. Secure Credential Management

The OpenAI credential is configured through n8n credential management. The API key should not be hard-coded into workflow nodes, exported JSON, screenshots, or the public GitHub repository. The exported workflow references the credential by name rather than exposing the secret.

4. Chat Model Configuration

Provider: OpenAI

Model: GPT-4.1

Temperature: 0.0

Max Tokens: 500

5. Test Execution

Required test prompt:

Hello, please introduce yourself in exactly one sentence.

The workflow successfully returned a response containing exactly one sentence. The execution was reviewed in n8n and showed successful execution with approximately 43 total tokens for the recorded test.

6. Execution Validation

The execution/debugger view was used to verify the workflow lifecycle and inspect execution output and token usage. The successful execution provides evidence that the trigger, AI Agent, and connected chat model operated correctly.

7. Submission Evidence

Deliverable	Status
Loom/YouTube demo video	To submit
Public GitHub repository	To submit
Workflow JSON export	Completed
Workflow screenshots	Completed

Workflow documentation PDF	Completed
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8. Repository Safety Note

Before making the GitHub repository public, verify that no API key, token, password, webhook secret, or other private credential is included in the workflow export, screenshots, or documentation.

9. Workflow Status

The latest exported workflow indicates **active: true**, confirming that the workflow is active in the current n8n configuration.